

METAL SOLUTIONS

EOS ToolSteel CM55

Material Data Sheet

EOS TOOLSTEEL CM55

EOS ToolSteel CM55 is a cobalt free ultra high strength steel for tooling and engineering solutions. Its alloying elements and moderate carbon content form a strong and stable structure for demanding applications. The properties of this steel make it suitable for cold- and hot-working tools. Producing mechanical engineering and powertrain components is well suited for this steel.

MAIN CHARACTERISTICS

- Cobalt free
- High strength and hardness
- Stable for use in elevated temperatures

Download Process Data Sheet (PDF) →

TYPICAL APPLICATIONS

- Cold working tools
- Hot working tools
- Parts for mechanical engineering
- Powertrain components

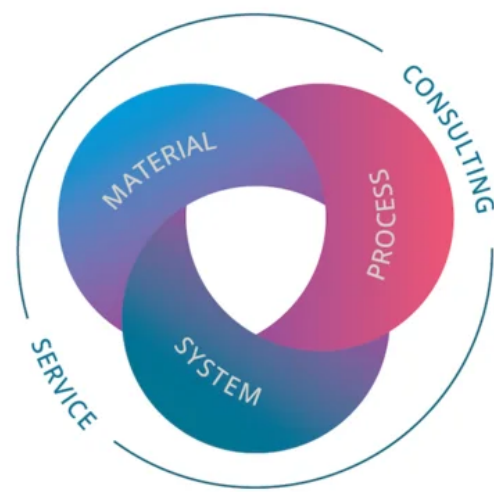
The EOS Quality Triangle

EOS uses an approach that is unique in the AM industry, taking each of the three central technical elements of the production process into account: the system, the material and the process. The data resulting from each combination is assigned a Technology Readiness Level (TRL) which makes the expected performance and production capability of the solution transparent.

EOS incorporates these TRLs into the following two categories:

- Premium products (TRL 7-9): offer highly validated data, proven capability and reproducible part properties.
- Core products (TRL 3 and 5): enable early customer access to newest technology still under development and are therefore less mature with less data.

All of the data stated in this material data sheet is produced according to EOS Quality Management System and international standards



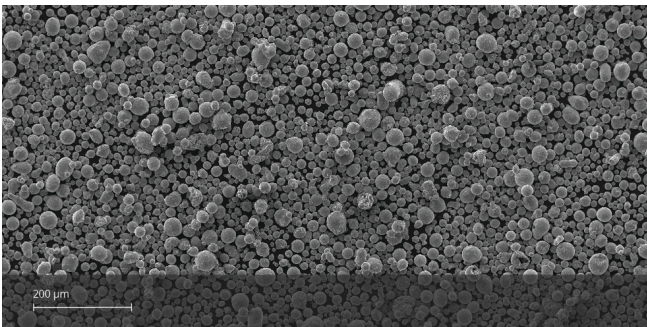
POWDER PROPERTIES

Powder Chemical Composition (wt.-%)

Element	Min.	Max.
Fe		Balance
Cr	5	8
Ni	5	8
Mo	0.5	1.2
Al	2	2.6
V	0.1	0.25
C	0.1	0.25

Powder Particle Size

Generic Particle Size Distribution	15 - 63 µm
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SEM micrograph of EOS ToolSteel CM55 powder

HEAT TREATMENT

Description

EOS ToolSteel CM55 can be heat treated to different hardness levels by adjusting the aging temperature. Final state of use is achieved by a two step heat treatment. First step is hardening where a strong martensitic microstructure is formed. The final hardness and strength is obtained in an aging treatment where strengthening phases and precipitates form.

Steps

1. Hardening

2 hours in 950 °C measured from the part when thoroughly heated through. Rapid gas cooling or quenching in oil. Cooling to room temperature before aging treatment.

2. Aging

4 hours in 550 °C - 630 °C depending on the desired hardness. Hold time when parts have thoroughly heated through. Air cooling or equivalent cooling rate. Peak strength and hardness is achieved by aging at 550 °C, stated data in this document represents this state. For lower hardness and strength choose aging temperature according to the graph below.

EOS ToolSteel CM55 for EOS M 290 | 40/80 µm

EOS M 290 - 40/80 µm - TRL 3

Process Information Metal

The parameter set has three different layer thickness options that can all be utilized within the same build: 40µm, 80 µm and 40/80 µm Skin. For high productivity needs a 80 µm parameter set is included . Both can be used separately for different parts or combined by using 40/80 µm Skin with faster 80 µm for the core of the part and higher quality 40 µm building for the surface of the part with defined thickness. Sectionioning parts in vertical direction for different parameters is also possible.

System Setup	EOS M 290
EOS Material set	CM55_Ar_040_080_CoreM291; CM55_N2_040_080_CoreM291
Software Requirements	EOSPRINT 2.10 or newer EOSYSTEM 2.14 or newer
Recoater Blade	Ceramic
Nozzle	EOS Grid Nozzle
Inert gas	Argon, Nitrogen
Sieve	75 µm

Additional Information	
Layer Thickness	40/80 µm
Volume Rate	4.1 - 7.8 mm³/s
Typical Dimensional Change after HT [%]	+0.2 %

Chemical and Physical Properties of Parts

Chemical composition of printed parts matches the chemistry of EOS ToolSteel CM55 powder.



Micrograph etched, heat treated state. Etchant: Oxalic acid

Microstructure of the Produced Parts

Defects	Thickness	Result	Number of Samples
Average Defect Percentage	40 μm	0.07 %	-
Average Defect Percentage	80 μm	0.12 %	-

Density EN ISO 3369	Thickness	Result	Number of Samples
Average Density	-	≥ 7.5 g/cm ³	-

Mechanical Properties

Mechanical Properties Heat Treated

EN ISO 6892-1 Room Temperature 40 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	1730	2030	4	-	230	-
Horizontal	1740	2040	4	-	230	-

Heat treated to 55 HRC.
Modulus of elasticity testing according to EN ISO 6892-1 Method A, Range 1 (0.00007 1/s)

Mechanical Properties Heat Treated

EN ISO 6892-1 Room Temperature 80 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	1700	2010	2.5	-	-	-
Horizontal	1710	2020	2.5	-	-	-

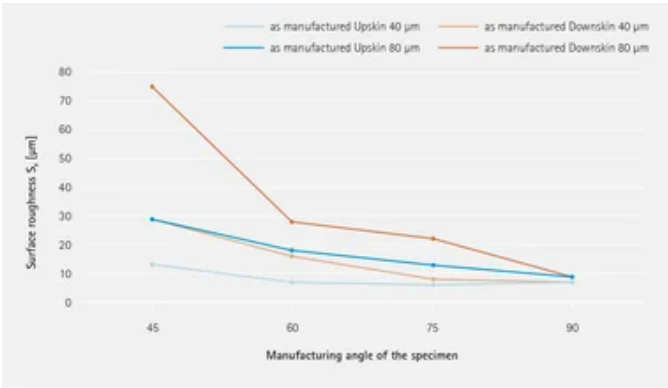
Heat treated to 55 HRC.
Modulus of elasticity testing according to EN ISO 6892-1 Method A, Range 1 (0.00007 1/s)

Impact Toughness

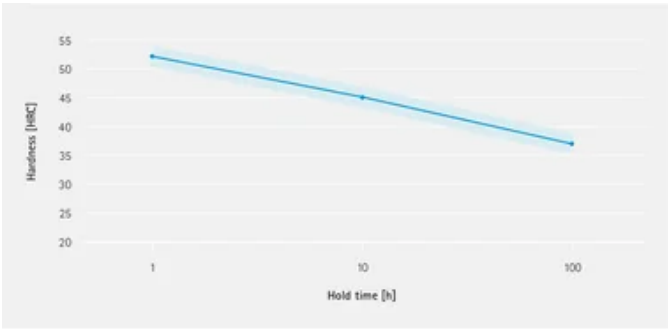
EN ISO 148-1, Charpy-V Heat Treated	
Typical Impact Toughness [J]	5

Testing according to ISO 148-1, V-notch at room temperature.

Surface Roughness



Temper resistance, 600°C



Coefficient of Thermal Expansion

ASTM E228	Temperature
10.6*10 ⁻⁶ /K	25 – 100 °C
11.5*10 ⁻⁶ /K	25 – 200 °C
12*10 ⁻⁶ /K	25 – 300 °C
12.4*10 ⁻⁶ /K	25 – 400 °C
12.7*10 ⁻⁶ /K	25 – 500 °C
13.2*10 ⁻⁶ /K	25 – 600 °C

HEADQUARTERS

EOS GmbH
Electro Optical Systems

Robert-Stirling-Ring 1
82152 Krailling / Munich
Germany

Tel.: +49 89 893 36-0
Email: info@eos.info
URL: www.eos.info

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